BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2023

MATHEMATICS

Past Paper 2023 • Paper Code: —

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Objective Type

Time Allowed: 20 Min. | **Maximum Marks:** 15

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Product of [x y][2; −1] is----------:

(A) [2x + y]
(B) [x − 2y]
(C) [2x − y]
(D) [x + 2y]

Q2. Which set has closure property w.r.t. addition:

(A) {0}
(B) {0, −1}
(C) {0, 1}
(D) {1/√2, 1/2}

Q3. log e = ---------, where e ≈ 2.718:

(A) 0
(B) 0.4343
(C) [illegible in source scan]
(D) 1

Q4. (√a + √b)(√a − √b) is equal to:

(A) a^2+b^2
(B) a^2-b^2
(C) a+b
(D) (a−b)

Q5. Factors of $5x^2 - 17xy - 12y^2$ are:

(A) (x+4y), (5x+3y)
(B) (x−4y), (5x−3y)
(C) (x−4y), (5x+3y)
(D) (5x−4y), (x+3y)

Q6. L.C.M. of a^2+b^2 and a^4-b^4 is:

(A) a^2+b^2 (B) a^4-b^4

(C) a^2-b^2 (D) $a-b$

Q7. Which is the solution of the inequality $3 - 4x \leq 11$:

(A) -8 (B) -2

(C) $-14/4$ (D) None of these

Q8. If (x, 0) = (0, y), then (x, y) is :

(A) (0, 0) (B) (0, 1)

(C) (1, 0) (D) (1, 1)

Q9. Distance between the points (1, 0) and (0, 10) is:

(A) 0 (B) 1

(C) $\sqrt{2}$ (D) 2

Q10. Sum of interior angles of a triangle is:

(A) 90° (B) 180°

(C) 360° (D) 0°

Q11. Total number of medians in a triangle are:

(A) 3 (B) 2

(C) 1 (D) 4

Q12. The right bisectors of the sides of the triangle are ----------:

(A) Parallel (B) Collinear

(C) Concurrent (D) Non-concurrent

Q13. Ratio between two elements 'a' and 'b' is represented by:

(A) a + b (B) a × b

(C) a − b (D) a : b

Q14. The region enclosed by the bounding lines of a closed figure is called ---------- of the figure:

(A) Perimeter (B) Area

(C) Volume (D) None of these

Q15. The medians of a triangle cut each other in the ratio ----------:

(A) 4 : 1 (B) 3 : 1

(C) 2 : 1 (D) 1 : 1

Essay Type — PART – I

Time Allowed: 2.10 hrs. | **Maximum Marks:** 60

Q2. (Marks 12) Write short answers to any SIX (6) questions:

(i) What is meant by adjoint of a matrix?

(ii) If A = [2 3; 1 0] and B = [5 −4; −2 −1] then find 2A + 3B

(iii) Simplify: $4(3)^n / (3^{n+1} - 3^n)$

(iv) Simplify: $\sqrt[4]{(81y^{-12}x^{-8})}$

(v) Calculate: $\log_3 2 \times \log_2 81$

(vi) Find the value of x: log x = 0.0044

(vii) If a + b = 10 and a − b = 6 then find the value of $a^2 + b^2$

(viii) Simplify: $\sqrt[5]{(243x^5y^{10}z^{15})}$

(ix) Factorize: $125x^3 - 216y^3$

Q3. (Marks 12) Write short answers to any SIX (6) questions:

(i) Use factorization to find the square root of: $4x^2 - 12x + 9$

(ii) Solve the inequality: $4x - 10.3 \leq 21x - 1.8$

(iii) Define strict inequalities.

(iv) Write the given equation in the form of y = mx + c: 3 − 2x + y = 0

(v) Define cartesian plane.

(vi) Find the distance between two points: A(−4, √2), B(−4, −3)

(vii) Define right angle triangle.

(viii) What is meant by H.S ≅ H.S ?

(ix) One exterior angle formed on producing one side of a parallelogram is 40°. Find the measures of its interior angles.

Q4. (Marks 12) Write short answers to any SIX (6) questions:

(i) Define bisection of an angle.

(ii) Justify that 2 cm, 3 cm and 5 cm are not the lengths of triangle.

(iii) Define ratio.

(iv) Define similar triangles.

(v) Find the value of x: [Figure: right triangle with vertical side x, base 1 cm, hypotenuse √2 cm, right-angle mark]

(vi) Find the area of given figure: [Figure: square 4 cm × 4 cm with right-angle mark]

(vii) Verify that the triangle having the following measures of sides are right-angled: a = 5cm, b = 12cm, c = [illegible in source scan]

(viii) Define ortho centre.

(ix) Construct a ΔABC in which: m AB = 4.8 cm, m BC = 3.7 cm, m∠B = 60°

> Note: Attempt any THREE questions in all. But question No. 9 is Compulsory. (PART-II)

Q5.

(a) [4] Solve the equation with the help of matrix inverse method: $3x - 4y = 4$, $x + 2y = 8$

(b) [4] Simplify: $(a^{2l}/a^{l+m})(a^{2m}/a^{m+n})(a^{2n}/a^{n+l})$

Q6.

(a) [4] Use logarithm table to find the value of: $\sqrt[5]{2.709} \times \sqrt[3]{1.239}$

(b) [4] If $a + b + c = 7$ and $ab + bc + ca = 9$ then find the value of $a^2 + b^2 + c^2$

Q7.

(a) [4] Factorize by factor theorem: $x^3 - x^2 - 22x + 40$

(b) [4] Simplify to the lowest form: $(x^3 - 8)/(x^2 - 4) \times (x^2 + 6x + 8)/(x^2 - 2x + 1)$

Q8.

(a) [4] Solve the inequalities: $-6 < (x - 2)/4 < 6$

(b) Construct ΔABC, draw perpendicular bisectors of its sides and verify that they are concurrent: m AB = 4cm, m BC = 4.8cm, m AC = 3.6 cm

Q9. Prove that "the right bisectors of the sides of a triangle are concurrent" OR Prove that "triangles on equal bases and of equal altitudes are equal in area".

GROUP – II

Objective Type

Time Allowed: 20 Min.	**Maximum Marks:** 15

> Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. $4x + 3y - 2$ is an algebraic ----------:

(A) Expression
(B) Sentence
(C) Equation
(D) Inequation

Q2. The symbol used for "is perpendicular to":

(A) $\parallel$
(B) $\perp$
(C) $\cong$
(D) $\leftrightarrow$

Q3. Congruent figures have same ----------:

(A) Mode
(B) Area
(C) Median
(D) Diagonal

Q4. Bisection means to divide into--------- equal parts:

(A) 3
(B) 1
(C) 2
(D) 4

Q5. Imaginary part of $-i(3i + 2)$ is ----------:

(A) 2ab
(B) -2
(C) 3
(D) -3

Q6. In parallelogram opposite sides are ----------:

(A) Opposite
(B) Diagonal
(C) Same
(D) Congruent

Q7. One angle on the base of an isosceles triangle is 30°, what is the measure of its vertical angle:

(A) 30°
(B) 60°
(C) 90°
(D) 120°

Q8. The logarithm of unity to any base is ----------:

(A) 1
(B) 10
(C) e
(D) 0

Q9. Factors of $5x^2 - 17xy - 12y^2$ are:

(A) (x+4y), (5x+3y)
(B) (x−4y), (5x−3y)
(C) (x−4y), (5x+3y)
(D) (5x−4y), (x+3y)

Q10. [√2 0; 0 √2] is called ---------- matrix:

(A) Zero
(B) Unit
(C) Scalar
(D) Singular

Q11. Mid point of the points (2, -2) and (-2, 2) is:

(A) (0, 0) (B) (1, 0)

(C) (0, 1) (D) (1, 1)

Q12. ---------- has no unit:

(A) Ratio (B) Proportion

(C) Congruent (D) Equality

Q13. If the capacity 'c' of an elevator is at most 1600 pounds, then ----------:

(A) $c < 1600$ (B) $c \geq 1600$

(C) $c \leq 1600$ (D) $c > 1600$

Q14. H.C.F of $a^3 + b^3$ and $a^2 - ab + b^2$ is ----------:

(A) $a + b$ (B) $a^2 + b^2$

(C) $(a - b)^2$ (D) $a^2 - ab + b^2$

Q15. If $y = 2x + 1$, $x = 2$ then y is ----------:

(A) 2 (B) 3

(C) 4 (D) 5

Essay Type — PART – I

Time Allowed: 2.10 hrs. | **Maximum Marks:** 60

Q2. (Marks 12) Write short answers to any SIX (6) questions:

(i) Define singular matrix.

(ii) If $A = [3\ 0; -1\ 2]$ and $B = [6; 5]$ then find AB.

(iii) Simplify: $\sqrt[3]{(-8/27)}$

(iv) Simplify and write answer in a + bi form: $-2/(1+i)$

(v) Find the value of x: $\log_3 x = 4$

(vi) Find the value of x: $\log x = 0.1821$

(vii) Reduce to the lowest form: $8a(x + 1) / 2(x^2 - 1)$

(viii) Simplify: $\sqrt[4]{(3/125)}$

(ix) Factorize: $x^2 - 11x - 42$

Q3. (Marks 12) Write short answers to any SIX (6) questions:

(i) Find the L.C.M.: $39x^7y^3z$, $91x^5y^6z^7$

(ii) Define linear inequality and write the standard form.

(iii) Solve: $-1/3\ x + 5 \leq 1$

(iv) Draw the point (4, -5) on the graph paper.

(v) Define cartesian plane.

(vi) Find the distance between the points: A(3, −11), B(3, −4)

(vii) Define equilateral triangle.

(viii) What is meant by S.S.S ≅ S.S.S?

(ix) Define parallelogram.

Q4. (Marks 12) Write short answers to any SIX (6) questions:

(i) Define the bisector of line segment.

(ii) 3 cm, 4 cm and 7 cm are not the lengths of the triangle. Give the reason.

(iii) Define ratio.

(iv) Define Pythagoras theorem.

(v) Find the value of x: [Figure: right triangle with vertical side 4 cm, base 3 cm, hypotenuse x]

(vi) Define altitude of a triangle.

(vii) Find the area of given figure: [Figure: large rectangle with a smaller square of side 4 cm at the bottom-left corner]

(viii) Define incentre of a triangle.

(ix) Construct a ΔXYZ in which mXY = 5.5 cm, mZX = 4.5 cm, m∠Z = 90°

> Note: Attempt any THREE questions in all. But question No. 9 is Compulsory. (PART-II)

Q5.

(a) Use the matrix inversion method to solve liner equations: $4x + 2y = 8;\ 3x - y = -1$

(b) Simplify: $(81)^n \times 3^5 - (3)^{4n-1}(243) / (9^{2n})(3^3)$

Q6.

(a) Use log tables to find the value of: 0.8176×13.64

(b) If $x^2 + y^2 + z^2 = 78$ and $xy + yz + zx = 59$, then find the value of $x + y + z$.

Q7.

(a) Factorize by factor theorem: $x^3 - 2x^2 - x + 2$

(b) Find the square root by division method: $x^4 - 10x^3 + 37x^2 - 60x + 36$

Q8.

(a) Solve: $|(x + 5)/(2 - x)| = 6$

(b) Construct triangle ABC, draw the perpendicular bisectors of its sides: mAB = 5.3 cm, m∠A = 45°, m∠B = 30°

Q9. Prove that any point on the right bisector of a line segment is equidistant from its end points. OR Prove that the parallelograms on the same base and between the same parallel lines (or of the same altitude) are equal in area.